

ESP32-S3 Microphone & RGB LED Ring PCB

Brief Design & Build Notes

Prepared by: EL Amraoui Hamza – @hamzaePCB
Project: Six-Microphone PDM Array and RGB Status Ring
Revision: Final Design
PCB CAD: KiCad

1 Purpose and Architecture

This PCB is an annular daughterboard designed for integration around the loudspeaker of an existing ESP32-S3 PoE speaker platform.

The ring integrates:

- Six digital PDM MEMS microphones.
- Six low-current RGB status LEDs.
- One LP5018 18-channel I²C constant-current LED driver.
- Local decoupling and supporting passive components.
- One 2×7 connector providing power and digital communication with the main ESP32-S3 board.

No additional microcontroller is required on the ring. Microphone acquisition and LED control are handled by the ESP32-S3 on the main board.

2 Main Board Interface

The ring is powered directly from the existing 3.3V rail through J1/J4. The interface is summarized below.

Pin	Signal
1	3V3
2	GND
3	PDM_CLK
4	GND
5	PDM_DATA0
6	GND
7	PDM_DATA1
8	GND
9	PDM_DATA2
10	GND
11	Reserved / Not Used
12	I2C_SDA
13	I2C_SCL
14	RING_GPIO_SPARE

Interleaved ground connections are used between the main PDM signals to provide convenient return paths through the board-to-board/cable interface.

3 PDM Microphone Architecture

3.1 Microphone Selection

The selected microphone is:

Infineon IM69D128SV01XTMA1

This device was selected because it provides a digital PDM interface compatible with the 3.3 V ring architecture and supports stereo PDM operation, where two microphones can share one data line by operating on opposite clock edges.

The microphone supply range includes the system 3.3 V rail, avoiding the need for an additional local microphone regulator.

3.2 Six-Microphone Bus Organization

The six microphones are organized as three PDM stereo pairs:

PDM Bus	Microphones	Configuration
PDM_DATA0	MIC1 + MIC2	Left / Right pair
PDM_DATA1	MIC3 + MIC4	Left / Right pair
PDM_DATA2	MIC5 + MIC6	Left / Right pair

All six microphones share the same PDM clock.

This architecture reduces the required number of ESP32-S3 data connections while maintaining six individual microphone channels.

PDM_DATA3 is not required by the final ring implementation and is therefore not populated as an active microphone bus.

4 Microphone Supply Decoupling

Each microphone includes a local bypass capacitor located close to its supply pins.

The purpose of this capacitor is to provide a short local path for transient current and to reduce high-frequency supply noise reaching the microphone.

Additional local bulk decoupling is provided on the ring supply near the active electronics.

The decoupling components should be placed with short connections to both 3V3 and the ground plane.

5 RGB LED Architecture

5.1 LED Driver Selection

The RGB status system requires:

$$6 \text{ RGB LEDs} \times 3 \text{ channels/LED} = 18 \text{ channels}$$

The selected driver is:

Texas Instruments LP5018RSMR

The LP5018 was selected because it provides exactly 18 constant-current LED sink channels and is controlled through I²C.

This provides several advantages over driving the LEDs directly from the ESP32-S3:

- Only the I²C interface is required for LED control.
- LED current is controlled by the driver rather than GPIO drive strength.
- Independent PWM brightness control is available for each channel.
- The six RGB LEDs can be controlled without consuming 18 MCU GPIOs.

6 LED Current Programming

The LP5018 output-current reference is programmed by resistor R1.

The selected value is:

$$R_{\text{IREF}} = 36.7 \text{ k}\Omega$$

Using the LP5018 current-setting relationship,

$$I_{\text{LED}} \simeq \frac{K \times V_{\text{IREF}}}{R_{\text{IREF}}}$$

with approximately

$$K \simeq 105$$

and

$$V_{\text{IREF}} \simeq 0.7 \text{ V},$$

the programmed channel current is:

$$I_{\text{LED}} \simeq \frac{105 \times 0.7}{36.7 \times 10^3}$$

$$\boxed{I_{\text{LED}} \approx 2.0 \text{ mA/channel}}$$

This current was intentionally kept low because the LEDs are status indicators located behind the speaker grille and do not require high illumination power.

It also reduces:

- Ring power consumption.
- LED-driver dissipation.
- Supply-current transients.
- Unnecessary optical brightness close to the microphone array.

7 RGB LED Current Budget

There are 18 independently driven LED channels.

At the programmed current:

$$I_{\text{LED,total}} = 18 \times 2 \text{ mA}$$

therefore,

$$I_{\text{LED,total}} \approx 36 \text{ mA}$$

for the conservative condition where all 18 channels are simultaneously driven at full programmed current.

Considering the LP5018 channel-current tolerance of approximately $\pm 7\%$, an upper estimate is:

$$I_{\text{LED,total,max}} = 36 \times 1.07$$

$$I_{\text{LED,total,max}} \approx 38.5 \text{ mA}$$

This represents a conservative LED-output current condition; normal status patterns are expected to consume less current because all colors are not necessarily driven simultaneously at full brightness.

8 Microphone Current Budget

At a PDM clock around 3.072 MHz, the IM69D128S current consumption used for the design budget is approximately:

$$I_{\text{MIC,typ}} \approx 520 \text{ }\mu\text{A}$$

per microphone.

For six microphones:

$$I_{\text{MIC,total,typ}} = 6 \times 520 \text{ }\mu\text{A} = 3.12 \text{ mA}$$

Using approximately $650 \text{ }\mu\text{A}$ per microphone as a conservative maximum design value:

$$I_{\text{MIC,total,max}} = 6 \times 650 \text{ }\mu\text{A} = 3.90 \text{ mA}$$

9 Ring 3.3-V Current Estimate

Combining the conservative microphone and LED-output estimates gives:

$$I_{\text{ring}} \approx I_{\text{LED}} + I_{\text{MIC}} + I_{\text{driver/internal}}$$

Before the LP5018 internal operating-current allowance:

$$I_{\text{LED,max}} + I_{\text{MIC,max}} = 38.5 + 3.9$$

$$\approx 42.4 \text{ mA}$$

Additional allowance is required for the LED-driver internal circuitry, I²C pull-ups and small auxiliary loads.

For system-level power allocation, approximately

$$\boxed{60 \text{ mA @ } 3.3 \text{ V}}$$

is therefore reserved for the complete ring.

The corresponding allocated electrical power is:

$$P_{\text{ring}} = 3.3 \times 0.060$$

$$\boxed{P_{\text{ring}} \approx 0.20 \text{ W}}$$

This is a design allocation rather than an expected continuous operating consumption.

10 PCB Implementation

10.1 Stack-Up

The final PCB uses a conventional two-layer FR-4 construction:

Layers	2
Material	FR-4
Finished thickness	1.6 mm
Outer copper	35 μm / 1 oz
Controlled impedance	Not required

A 1.6 mm board thickness was retained to provide mechanical rigidity for the relatively large annular PCB.

10.2 Routing Strategy

The main routing strategy is:

- Approximately 0.50 mm for the main 3V3 distribution.
- Approximately 0.25 mm for PDM clock and PDM data signals.
- Approximately 0.30 mm for I²C routing where applicable.
- Local smaller traces for low-current LED-driver outputs where required.
- Ground copper planes on both sides of the PCB.
- Ground stitching vias distributed around the ring.

The PDM clock is routed over a continuous ground reference where practical. Unnecessary long stubs and extended parallel routing with switching LED signals are avoided.

No controlled-impedance requirement is imposed on the PDM or I²C buses for this design.

11 Assembly Notes

Before assembly, verify the following orientation-sensitive components:

- Pin-1 orientation of the LP5018.
- Pin-1 and acoustic-port orientation of each PDM microphone.
- RGB LED orientation/polarity.
- J1 connector pin-1 orientation.

The LP5018 exposed center pad must be soldered correctly to the PCB ground structure.

Microphone acoustic ports must remain unobstructed after assembly and after installation behind the speaker grille.

No solder, adhesive, conformal coating or mechanical material should block the microphone acoustic openings.

12 Manufacturing Package

The final delivery package contains:

- KiCad schematic and PCB source files.
- Gerber fabrication files.
- NC drill files.
- Bill of Materials (BOM).
- Component Placement List (CPL / Pick-and-Place).
- STEP mechanical model.
- Component datasheets / supporting documentation.
- This brief design and build note.

13 Final Build Summary

The final ring uses a 3.3V architecture shared with the existing ESP32-S3 main board. Six PDM microphones are arranged as three stereo data pairs, while the six RGB LEDs are controlled through one 18-channel LP5018 driver.

The LED current is intentionally limited to approximately 2mA per channel using a 36.7k Ω IREF resistor. The complete ring is allocated approximately 60mA from the 3.3V rail for system-level power budgeting.

The PCB is implemented as a two-layer, 1.6mm, 1-oz FR-4 board with ground-plane referencing and local supply decoupling.